Introduction to Internet of Things (IoT)

A simple reference document for testing a Retrieval-Augmented Generation (RAG) document bot.

1. What is IoT?

The Internet of Things (IoT) is a system in which physical devices are connected to a network so that they can collect, exchange, and sometimes process data. IoT devices commonly use sensors, controllers, communication networks, and software.

2. Main Components of an IoT System

Component	Purpose
Sensor	Measures physical conditions such as temperature, light, pressure, or motion.
Controller	Processes sensor data and controls connected devices.
Communication Network	Transfers data between devices and other systems.
Cloud / Server	Stores and processes data and may provide remote access.
Actuator	Performs a physical action based on a control command.

3. How an IoT System Works

A typical IoT system follows a simple sequence: a sensor collects data, a controller processes the data, a communication network transfers the information, and a server or cloud platform can store and analyze it. If an action is required, the controller sends a command to an actuator.

4. Example: Smart Temperature Monitoring

A temperature sensor can measure the temperature of a room. A microcontroller reads the sensor value and sends it through Wi-Fi to a server. The server stores the temperature data. A user can view the temperature using a dashboard. If the temperature becomes too high, the system can activate a cooling device through an actuator.

5. Common IoT Communication Technologies

Common communication technologies used in IoT include Wi-Fi, Bluetooth, Zigbee, LoRa, and cellular networks. The choice depends on factors such as communication range, power consumption, data rate, and application requirements.

6. Advantages of IoT

• Remote monitoring of physical systems.

• Automatic collection of sensor data.

• Automation of repetitive tasks.

• Real-time access to useful information.

• Improved control and resource management.

7. Challenges of IoT

• Security and privacy of connected devices.

• Reliable network connectivity.

• Power consumption in battery-operated devices.

• Managing large amounts of sensor data.

• Compatibility between different devices and platforms.

8. Key Terms

Term	Simple Meaning
IoT	Network of connected physical devices.
Sensor	Device that measures a physical quantity.
Actuator	Device that performs a physical action.
Microcontroller	Small computer used to control electronic systems.
Cloud	Remote computing and storage infrastructure accessed through a network.

RAG Testing Tip: This document is intentionally simple and contains clear facts, definitions, component descriptions, and one example. It is suitable for testing questions such as "What is IoT?", "What does a sensor do?", "Name three IoT communication technologies", "What are two challenges of IoT?", and "How does the smart temperature monitoring example work?"